Visual and Ocular Reflexes

Pupillary Light Reflex: Neural Pathway

- **Function:** Regulates light entering the eye through pupillary constriction (miosis)
- **Stimulus:** Light entering the retina
- **Afferent limb (CN II):** Retina → optic nerve → pretectal nuclei (midbrain)
- **Central connection**: Pretectal nuclei → bilateral Edinger–Westphal nuclei
- **Efferent limb (CN III):** Edinger–Westphal nucleus → CN III → ciliary ganglion → short ciliary nerves → sphincter pupillae
- **Response:** Direct and consensual pupillary constriction

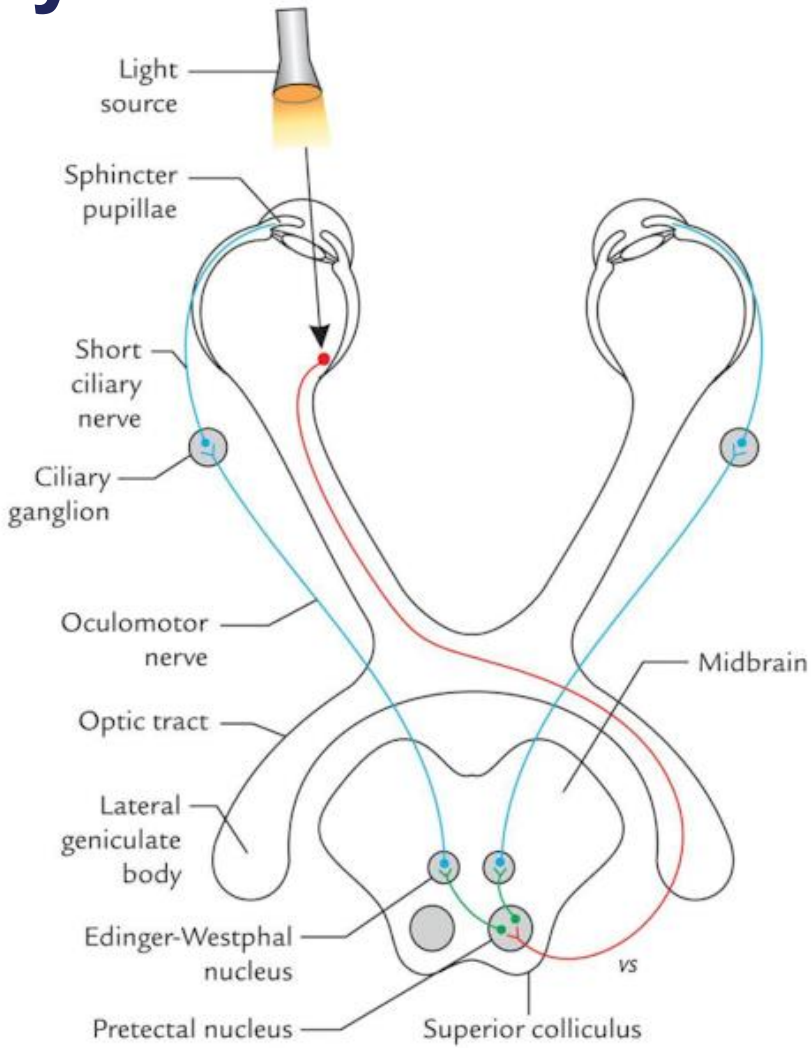


FIG. 18.11 Pathways for light reflex (direct and consensual).

Pupillary Light Reflex: CN II vs CN III Lesions

Lesion	Light in Affected Eye	Light in Unaffected Eye
CN II (afferent)	Neither pupil constricts	Both pupils constrict
CN III (efferent)	Only unaffected pupil constricts	Only unaffected pupil constricts

CN II lesion → defective sensory input

CN III lesion → defective pupillary constriction

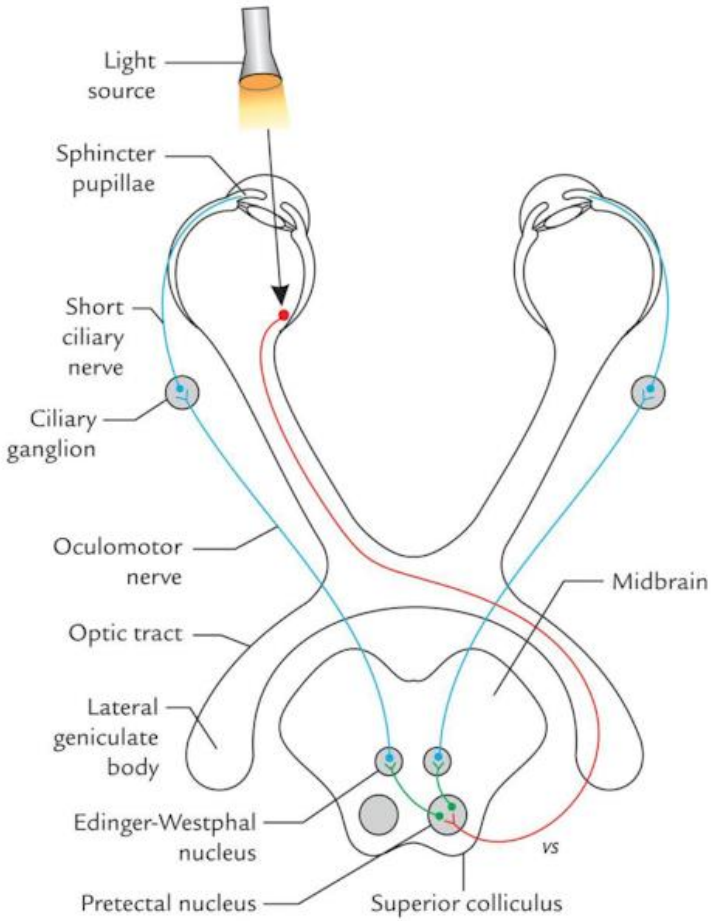


FIG. 18.11 Pathways for light reflex (direct and consensual).

Relative Afferent Pupillary Defect (Marcus Gunn Pupil)

- **Cause:** Reduced sensory input from one eye, commonly due to optic nerve disease
- **Test:** Swinging flashlight test
- Light in healthy eye → both pupils constrict
- Light moved to affected eye → both pupils dilate relatively (reduced constriction)
- Efferent CN III pathways remain intact

RAPD → Afferent defect (CN II), not an efferent CN III lesion

Near Response (Accommodation–Convergence)

The near response consists of three coordinated changes when shifting focus from a distant to a near object.

1. **Accommodation:** Ciliary muscle contracts → zonular fibers relax → lens becomes rounder
2. **Convergence:** Medial rectus muscles contract (CN III) → eyes turn inward
3. **Pupillary Constriction (Miosis):** Sphincter pupillae contracts → reduces pupil diameter

Near response = Accommodation + Convergence + Miosis

Neural Pathways of the Near Response

Afferent Pathway

- Retina → CN II → visual cortex → supranuclear near-response pathways

Efferent Pathways

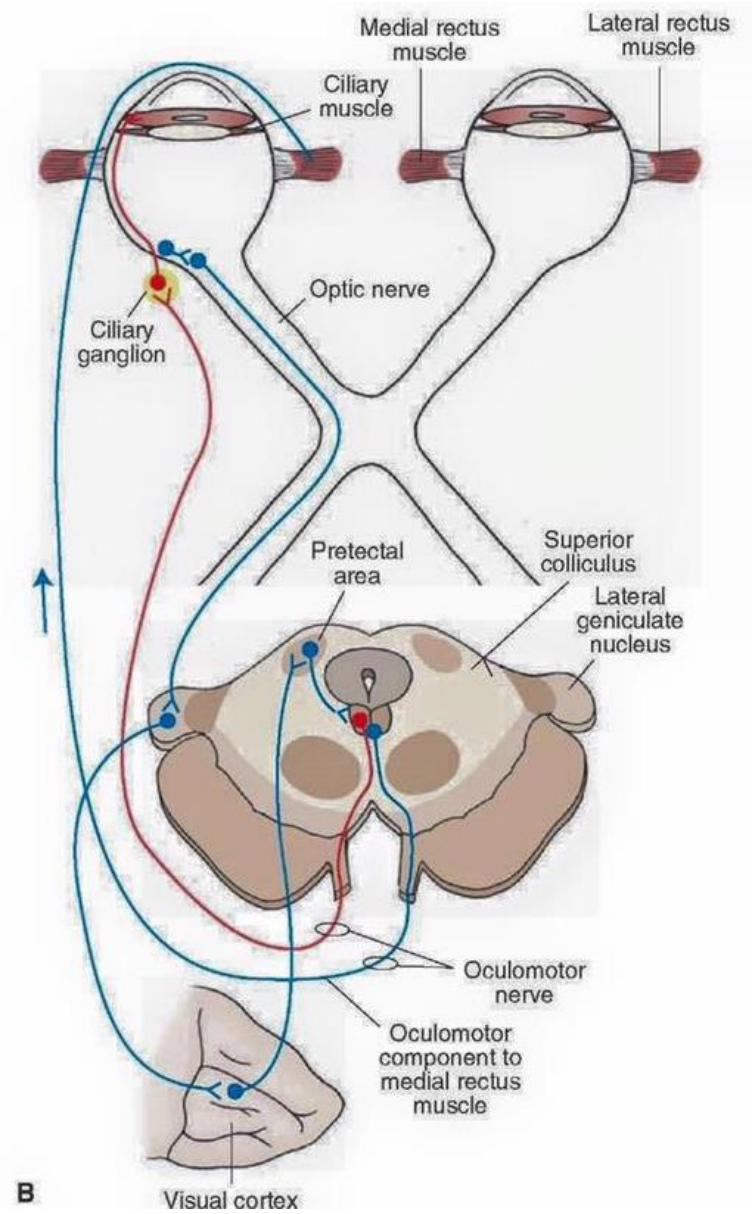
Parasympathetic (GVE)

- Edinger–Westphal nucleus → CN III → ciliary ganglion
  - Ciliary muscle → accommodation
  - Sphincter pupillae → miosis

Somatic Motor (GSE)

- Oculomotor nucleus → CN III → medial rectus muscles
- Bilateral eye convergence

CN III coordinates all three components of the near response



Light-Near Dissociation

- Pupillary response to light is impaired
- Pupillary constriction during near fixation is preserved or relatively better

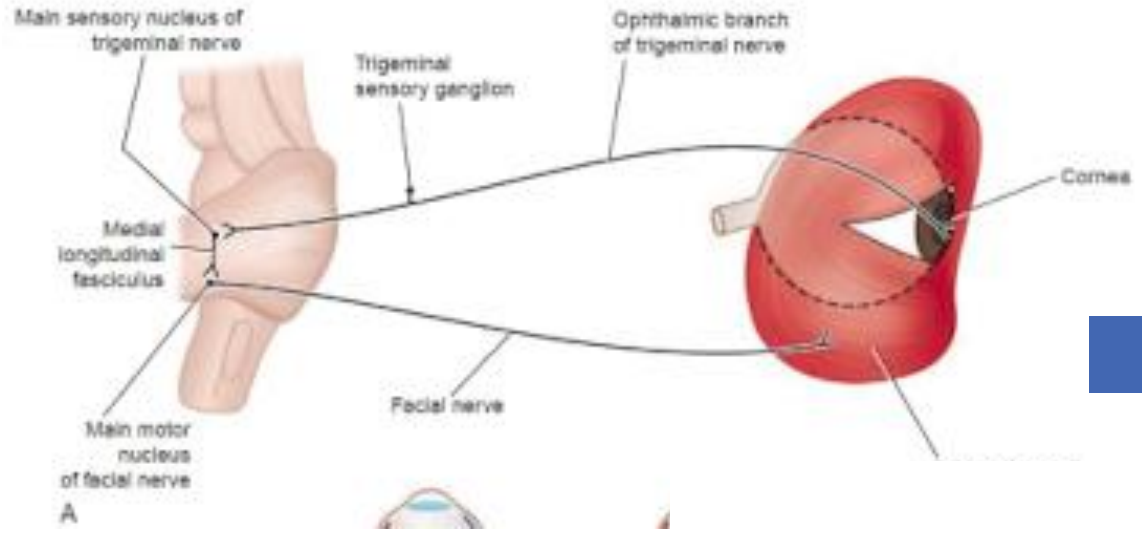
Clinical Associations

- Argyll Robertson pupil: small, irregular pupils; classically associated with neurosyphilis
- Adie's tonic pupil: dilated pupil with slow, sustained near constriction
- Parinaud syndrome: dorsal midbrain lesion disrupting pupillary light-reflex pathways

Light response impaired | Near constriction relatively preserved

Corneal (Blink) Reflex

- Function: Protective reflex producing bilateral eyelid closure
- Stimulus: Light touch of the cornea
- Afferent limb (GSA): CN V1 (nasociliary nerve) → spinal trigeminal nucleus
- Central connections: Trigeminal sensory pathways → bilateral facial motor nuclei
- Efferent limb (SVE): CN VII → orbicularis oculi
- Response: Bilateral blinking (direct + consensual)



Corneal reflex: V1 in → VII out → bilateral blinking

Corneal Reflex: Lesion Localization

Lesion	Touch affected cornea	Touch unaffected cornea
CN V1 (afferent)	Neither eye blinks	Both eyes blink
CN VII (efferent)	Only unaffected eye blinks	Only unaffected eye blinks

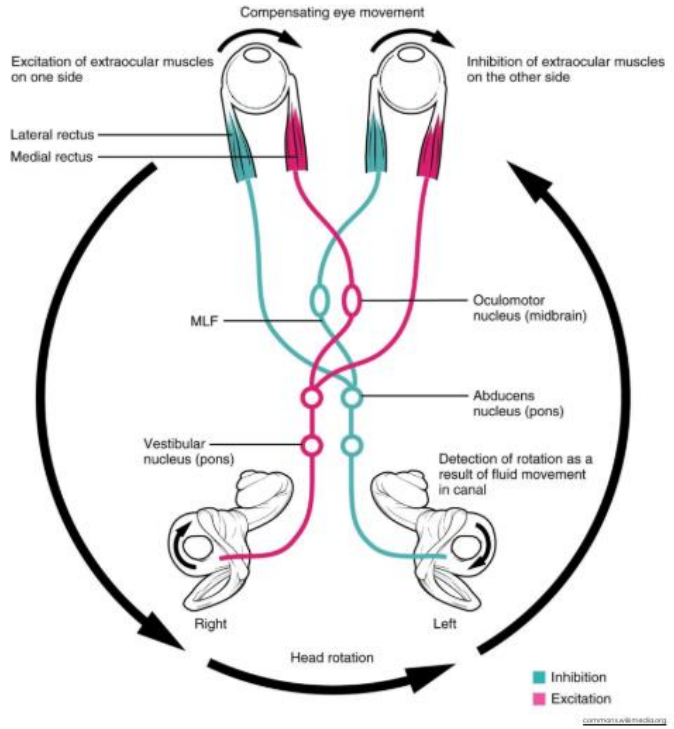
Clinical Correlation

- CN V1 lesion: loss of sensory input from affected cornea
- CN VII lesion: impaired ipsilateral eyelid closure (e.g., Bell palsy)

Vestibulo-Ocular Reflex (VOR)

- **Function:** Stabilizes gaze during head movement
- **Stimulus:** Head rotation detected by semicircular canals
- **Afferent limb:** Vestibular division of CN VIII
- **Central pathway:** Vestibular nuclei → CN VI nucleus → MLF → CN III nucleus
- **Efferent limb:** CN VI (lateral rectus) + CN III (medial rectus)
- **Response:** Eyes move opposite to head rotation

- Head turns one way → eyes move the opposite way
- VOR stabilizes the retinal image during head movement without requiring conscious visual processing.

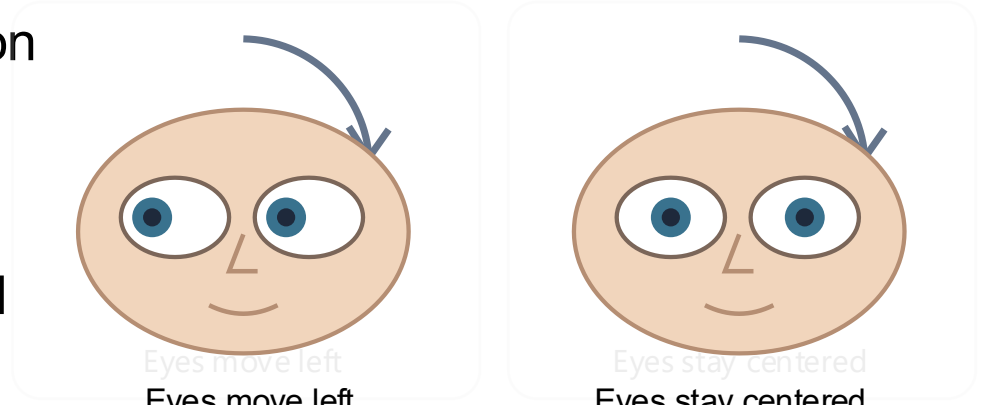


Oculocephalic Reflex (Doll's Eye Maneuver)

- Assesses vestibular–brainstem–ocular motor pathways in unconscious patients
- Intact VOR: Eyes move opposite to head rotation, maintaining gaze direction in space
- Absent VOR: Eyes remain stationary within the orbits and move with the head
- Abnormal response may indicate brainstem dysfunction

Clinical Precaution

- Do not perform if cervical spine instability is suspected



Eyes move left

Eyes stay centered

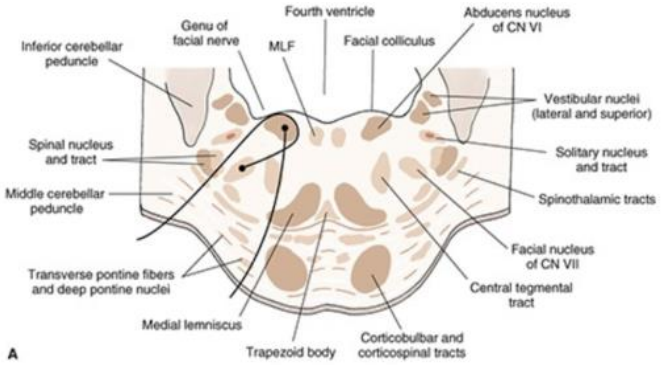
Schematic: in both examples the head turns right; the eye positions within the orbits differ.

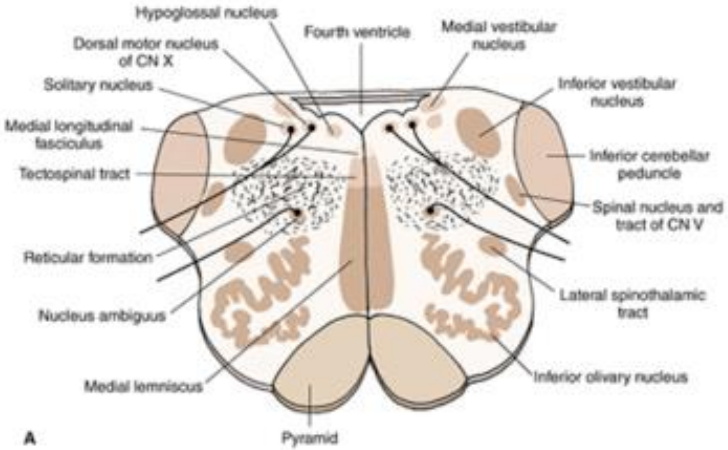
Oculocephalic reflex = clinical assessment of VOR integrity

Medial Longitudinal Fasciculus (MLF)

- Location: Paired white matter tract near the midline of the brainstem
- Connects vestibular nuclei and ocular motor nuclei (CN III, IV, VI)
- Coordinates conjugate eye movements
- Essential for the vestibulo-ocular reflex
- Lesion disrupts coordinated horizontal gaze

MLF → coordinates eye movements between CN III, IV, VI and vestibular nuclei





MLF spans from the upper medulla → pons → midbrain, running close to the midline in the tegmentum of each section. Siegel, EN, 4e.

Internuclear Ophthalmoplegia (INO)

- Cause: Lesion of the medial longitudinal fasciculus (MLF)
- Ipsilateral eye: Impaired adduction during horizontal gaze
- Contralateral eye: Abducting nystagmus
- Convergence: Usually preserved
- Clinical associations: Multiple sclerosis (younger patients), brainstem stroke (older patients)
- Example: Left MLF lesion → left eye fails to adduct on rightward gaze

MLF lesion → ipsilateral adduction deficit + contralateral abducting nystagmus

Summary

- Cornea and lens: Refract light and focus images on the retina
- Retina: Detects light; fovea provides highest visual acuity
- Visual pathway: Retina → CN II → chiasm → optic tract → LGN → radiations → V1
- Optic chiasm: Nasal fibers cross; each hemisphere processes the opposite visual field
- Visual field defects: Optic nerve → monocular; chiasm → bitemporal; postchiasmal → homonymous loss
- Pupillary light reflex: CN II → pretectal nuclei → Edinger–Westphal nuclei → CN III
- Near reflex: Accommodation + convergence + miosis
- Corneal reflex = V1 in → VII out
- VOR and MLF: Coordinate eye movements; MLF lesions cause internuclear ophthalmoplegia (INO)